

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 24

Design of IIR Filters from Analog Filters — Impulse Invariance

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of Impulse Invariance

1.1 Definition & Invariance Criterion

The **Impulse Invariance Method** designs a digital IIR filter by sampling the continuous-time impulse response $h_a(t)$ of an analog prototype filter at uniform sampling intervals T_s :

$$\mathbf{h}[n] = T_s \mathbf{h}_a(nT_s), \quad n = 0, 1, 2, \dots \quad (1)$$

The factor T_s scales the discrete response so that the digital filter's low-frequency gain matches the analog filter.

1.2 Mathematical Proof of the Pole Mapping Formula

Let the analog filter transfer function $H_a(s)$ have distinct poles s_k and be expanded into partial fractions:

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \quad (2)$$

We derive the digital filter transfer function $H(z)$ via three rigorous mathematical steps:

1. Step 1: Inverse Laplace Transform to continuous-time:

$$h_a(t) = \mathcal{L}^{-1}\{H_a(s)\} = \sum_{k=1}^N A_k e^{s_k t} u(t) \quad (3)$$

2. Step 2: Periodic Sampling with Scaling T_s ($t = nT_s$):

$$h[n] = T_s h_a(nT_s) = T_s \sum_{k=1}^N A_k e^{s_k nT_s} u[n] = \sum_{k=1}^N T_s A_k (e^{s_k T_s})^n u[n] \quad (4)$$

3. Step 3: Z-Transform via Infinite Geometric Series: Applying $\mathcal{Z}\{a^n u[n]\} = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}$ (convergent for $|z| > |a|$):

$$\begin{aligned} H(z) &= \sum_{n=0}^{\infty} h[n] z^{-n} = \sum_{n=0}^{\infty} \left[\sum_{k=1}^N T_s A_k (e^{s_k T_s})^n \right] z^{-n} \\ &= \sum_{k=1}^N T_s A_k \sum_{n=0}^{\infty} (e^{s_k T_s} z^{-1})^n = \sum_{k=1}^N \frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}} \end{aligned} \quad (5)$$

Therefore, each individual analog partial fraction term maps directly according to:

$$\frac{\mathbf{A}_k}{s - s_k} \xrightarrow{\text{Impulse Invariance}} \frac{T_s \mathbf{A}_k}{1 - e^{s_k T_s} z^{-1}} \quad (6)$$

Every continuous pole at $s = s_k$ maps directly to a discrete pole at:

$$\mathbf{z}_k = e^{s_k T_s} \quad (7)$$

2 Stability Mapping & The Aliasing Dilemma

2.1 Preservation of Stability

Let an analog pole be $s_k = \sigma_k + j\Omega_k$. The corresponding digital pole has magnitude:

$$|z_k| = |e^{(\sigma_k + j\Omega_k)T_s}| = e^{\sigma_k T_s} \quad (8)$$

- If $\sigma_k < 0$ (Left-Half s -Plane) $\implies |z_k| = e^{\sigma_k T_s} < 1$ (**Inside the Unit Circle**).
- **Theorem:** A stable analog filter is ****always mapped into a strictly stable digital filter****.

2.2 Frequency Relationship & The Aliasing Problem

The normalized digital frequency ω is related to analog frequency Ω by:

$$\omega = \Omega T_s \quad (9)$$

The discrete frequency spectrum is an infinite periodic superposition of shifted analog spectra:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega - 2\pi k}{T_s}\right) \quad (10)$$

Because the analog rational transfer function $H_a(j\Omega)$ cannot be strictly band-limited to $|\Omega| \leq \pi/T_s$, the high-frequency tails fold back into the baseband $[-\pi, \pi]$, creating ****aliasing distortion****.

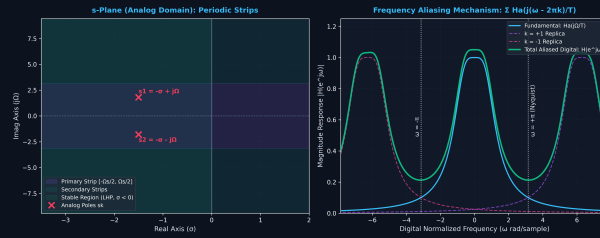


Figure 1: Left: Horizontal strips of width $2\pi/T_s$ in the s -plane mapping into the z -plane unit circle. Right: Aliasing caused by overlapping tails of the periodic spectrum copies.

- **Suitable Filters:** Lowpass and Bandpass filters whose analog stopband attenuates deeply (> 30 – 40 dB) at the Nyquist frequency $\Omega_N = \pi/T_s$.
- **Unsuitable Filters:** Highpass and Bandstop filters **CANNOT** be designed via impulse invariance because their high-frequency gains fold over into DC ($\omega = 0$), destroying stopband attenuation.

3 Step-by-Step Design Protocol

1. **Identify Sampling Interval T_s and Frequencies:** Convert digital specifications using $\Omega = \omega/T_s$.
2. **Factor the Analog Transfer Function:** Find all poles s_k of $H_a(s)$.
3. **Perform Partial Fraction Expansion:**

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k}, \quad \text{where } A_k = \lim_{s \rightarrow s_k} (s - s_k) H_a(s) \quad (11)$$

4. **Apply Pole Mapping Formula:** Replace each term by $\frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}}$.
5. **Combine Conjugate Terms:** Put complex conjugate pole pairs over a common quadratic denominator to yield real biquad filter coefficients.

4 Standard Solved Exam Example

Problem 1: An analog second-order Butterworth lowpass filter with cutoff frequency $\Omega_c = 1$ rad/s has the transfer function:

$$H_a(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad (12)$$

Design the corresponding digital IIR filter using the Impulse Invariance method with sampling period $T_s = 1$ second. Verify filter stability and calculate the DC gain.

Solution:

Step 1: Determine Analog Poles via Quadratic Formula Setting the denominator polynomial to zero:

$$s^2 + \sqrt{2}s + 1 = 0 \quad (13)$$

Applying the quadratic formula with all intermediate arithmetic steps:

$$\begin{aligned} s_{1,2} &= \frac{-\sqrt{2} \pm \sqrt{(\sqrt{2})^2 - 4(1)(1)}}{2(1)} = \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} \\ &= \frac{-\sqrt{2} \pm \sqrt{-2}}{2} = \frac{-\sqrt{2} \pm j\sqrt{2}}{2} = -\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}} \approx -\mathbf{0.7071} \pm \mathbf{j0.7071} \end{aligned} \quad (14)$$

Step 2: Partial Fraction Expansion and Residue Calculations Express $H_a(s)$ in partial fractions:

$$H_a(s) = \frac{1}{(s - s_1)(s - s_2)} = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} \quad (15)$$

Computing the residue A_1 using the Heaviside limit method:

$$\begin{aligned} A_1 &= \lim_{s \rightarrow s_1} (s - s_1)H_a(s) = \frac{1}{s - s_2} \Big|_{s=s_1} = \frac{1}{s_1 - s_2} \\ &= \frac{1}{\left(-\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right)} = \frac{1}{j\frac{2}{\sqrt{2}}} = \frac{1}{j\sqrt{2}} = -\mathbf{j}\frac{1}{\sqrt{2}} \approx -\mathbf{j0.7071} \end{aligned} \quad (16)$$

Since the poles are complex conjugates, the second residue is the conjugate of the first:

$$A_2 = A_1^* = +\mathbf{j}\frac{1}{\sqrt{2}} \approx +\mathbf{j0.7071} \quad (17)$$

Step 3: Map Analog Poles to Z-Plane Discrete Poles ($T_s = 1$ s) Using the mapping rule $z_k = e^{s_k T_s}$ with $T_s = 1$:

$$\begin{aligned} z_1 &= e^{s_1 \cdot 1} = e^{-0.7071 + j0.7071} = e^{-0.7071} \cdot e^{j0.7071} \\ &= e^{-0.7071} [\cos(0.7071 \text{ rad}) + j \sin(0.7071 \text{ rad})] \end{aligned} \quad (18)$$

Converting radians to degrees: $\theta = 0.7071 \times \frac{180^\circ}{\pi} \approx 40.514^\circ$. Evaluating values:

$$\begin{aligned} e^{-0.7071} &\approx 0.49307 \\ \cos(40.514^\circ) &\approx 0.76024, \quad \sin(40.514^\circ) \approx 0.64964 \\ z_1 &= 0.49307(0.76024 + j0.64964) = \mathbf{0.37485} + \mathbf{j0.32032} \approx \mathbf{0.3749} + \mathbf{j0.3203} \\ z_2 &= z_1^* = \mathbf{0.3749} - \mathbf{j0.3203} \end{aligned} \quad (19)$$

Stability check: $|z_1| = |z_2| = e^{\sigma_1 T_s} = e^{-0.7071} = 0.4931 < 1$. Both poles lie inside the unit circle; the filter is strictly stable.

Step 4: Algebraic Synthesis of the Real Second-Order Biquad Applying the impulse invariance transformation $\frac{A_k}{s - s_k} \rightarrow \frac{T_s A_k}{1 - z_k z^{-1}}$:

$$\begin{aligned} H(z) &= \frac{T_s A_1}{1 - z_1 z^{-1}} + \frac{T_s A_2}{1 - z_1^* z^{-1}} = \frac{-j\frac{1}{\sqrt{2}}(1 - z_1^* z^{-1}) + j\frac{1}{\sqrt{2}}(1 - z_1 z^{-1})}{(1 - z_1 z^{-1})(1 - z_1^* z^{-1})} \\ &= \frac{j\frac{1}{\sqrt{2}}(z_1^* - z_1)z^{-1}}{1 - (z_1 + z_1^*)z^{-1} + |z_1|^2 z^{-2}} = \frac{j\frac{1}{\sqrt{2}}(z_1^* - z_1)z^{-1}}{1 - 2\text{Re}(z_1)z^{-1} + |z_1|^2 z^{-2}} \end{aligned} \quad (20)$$

Computing each intermediate term explicitly:

$$\begin{aligned} z_1^* - z_1 &= -j2\text{Im}(z_1) = -j2(0.32032) = -j0.64064 \implies \text{Numerator} = j\frac{1}{\sqrt{2}}(-j0.64064) = \mathbf{0.4530} \\ 2\text{Re}(z_1) &= 2(0.37485) = \mathbf{0.7497} \approx \mathbf{0.7496}, \quad |z_1|^2 = (0.49307)^2 = \mathbf{0.2431} \end{aligned} \quad (21)$$

The final digital filter system transfer function is:

$$\mathbf{H(z)} = \frac{\mathbf{0.4530z^{-1}}}{\mathbf{1 - 0.7496z^{-1} + 0.2431z^{-2}}} \quad (22)$$

Step 5: DC Gain and Aliasing Analysis Evaluating $H(z)$ at DC ($z = e^{j0} = 1$):

$$H(1) = \frac{0.4530(1)}{1 - 0.7496(1) + 0.2431(1)} = \frac{0.4530}{0.4935} \approx \mathbf{0.9179} \quad (23)$$

Physical Meaning: The analog prototype has DC gain $H_a(0) = 1.0$. The digital DC gain is 0.9179. The difference (0.9179 vs 1.0000) is the exact quantitative manifestation of **aliasing distortion** caused by selecting a relatively large sampling period $T_s = 1$ s.

Solved Exam Example 2: First-Order Prototype & Impact of Sampling Period

Problem 2: An analog lowpass system is described by $H_a(s) = \frac{2}{s+2}$. Find the digital filter $H(z)$ using Impulse Invariance for: (a) $T_s = 0.1$ s, and (b) $T_s = 0.5$ s. Compare the DC gains.

Solution: The analog pole is at $s_1 = -2$, with residue $A_1 = 2$. By impulse invariance:

$$H(z) = \frac{T_s A_1}{1 - e^{s_1 T_s} z^{-1}} = \frac{2T_s}{1 - e^{-2T_s} z^{-1}} \quad (24)$$

$$1. \text{ Case (a): } T_s = 0.1 \text{ s: } e^{-2(0.1)} = e^{-0.2} \approx 0.8187 \implies H(z) = \frac{\mathbf{0.2}}{\mathbf{1 - 0.8187z^{-1}}}.$$

$$\text{DC Gain: } H(1) = \frac{0.2}{1 - 0.8187} = \frac{0.2}{0.1813} = \mathbf{1.103} \quad (\approx \text{analog gain } 1.0).$$

$$2. \text{ Case (b): } T_s = 0.5 \text{ s: } e^{-2(0.5)} = e^{-1.0} \approx 0.3679 \implies H(z) = \frac{\mathbf{1.0}}{\mathbf{1 - 0.3679z^{-1}}}.$$

$$\text{DC Gain: } H(1) = \frac{1.0}{1 - 0.3679} = \frac{1.0}{0.6321} = \mathbf{1.582} \quad (\text{severe aliasing error!}).$$

Conclusion: As T_s increases (lower sampling rate), aliased frequency copies overlap heavily into base-band, distorting filter gain. Smaller T_s significantly reduces aliasing.

5 Essential Exam Checklist & Traps

- ☐ **Partial Fraction Rule:** Always expand $H_a(s)$ into partial fractions first. Never substitute $s = \frac{1}{T} \ln(z)$ algebraically into $H_a(s)$.
- ☐ **The Scaling Factor T_s :** Always include T_s in the numerator: $\frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}}$.
- ☐ **Zero Locations:** Remember that ****zeros do not map**** via $z = e^{sT_s}$. Only poles obey this mapping.
- ☐ **Highpass Prohibition:** Impulse Invariance cannot be used to design Highpass filters due to infinite spectral foldover (aliasing) at DC.